

Tangent-Space Methods for Nonlinear Control and Biomechanics

Volume V: Practical Application

The Companion Implementation Platform

Preface

Draft Notice. This volume is under active development. All chapters are structural outlines with preliminary material that will be expanded in future editions. Exercises, worked examples, and backmatter (glossary, further reading) have not yet been written.

This is a book about *doing*. Volumes 0–IV developed the theory of motion: from differential geometry through nonlinear control, biomechanics, and motor control. This volume puts every tool to work.

The companion software developed alongside this textbook, UpstreamDrift, provides a pedagogical simulation environment for multi-body dynamics, trajectory optimization, and controller design. It wraps established physics engines (MuJoCo, Drake, Pinocchio, OpenSim) behind a consistent Python API, allowing the reader to run every example from the preceding four volumes and to build their own analyses.

Disclosure. UpstreamDrift is developed by the authors of this textbook. It is not a production simulation engine; it is an educational platform designed to accompany this text and make the theory from Volumes 0–IV executable. The underlying physics computations are delegated to the established engines listed above.

Every chapter in this volume produces a concrete, runnable result. There are no hypothetical exercises—every example uses real code, real models, and real data. By the end, you will have implemented a complete pipeline from model construction through trajectory optimization to robustness analysis, applied to the golf swing as the running case study.

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Nomenclature

General Conventions

- Scalars are lowercase or Greek letters (e.g., m, t, θ).
- Vectors are bold lowercase (e.g., $\mathbf{x}, \mathbf{u}, \mathbf{q}$).
- Matrices are bold uppercase (e.g., $\mathbf{A}, \mathbf{B}, \mathbf{M}, \mathbf{J}$).
- Expected values and sets are denoted by blackboard bold or script (e.g., $\mathbb{E}, \mathcal{S}$).

State and Kinematics

$\mathbf{q} \in \mathbb{R}^n$ Joint configuration vector (generalized coordinates).

$\dot{\mathbf{q}} \in \mathbb{R}^n$ Joint velocity vector (generalized velocities).

$\ddot{\mathbf{q}} \in \mathbb{R}^n$ Joint acceleration vector.

$\mathbf{x} \in \mathbb{R}^{n_x}$ System state vector; commonly $\mathbf{x} = [\mathbf{q}^T, \dot{\mathbf{q}}^T]^T$ for mechanical systems.

$\mathbf{u} \in \mathbb{R}^m$ Control input vector (e.g., joint torques, muscle activations).

$\boldsymbol{\tau} \in \mathbb{R}^n$ Generalized forces/torques acting on the joints.

$t \in \mathbb{R}$ Time.

Inertia and Dynamics

$\mathbf{M}(\mathbf{q})$ Mass (inertia) matrix, symmetric positive definite.

$\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})$ Coriolis and centrifugal force matrix.

$\mathbf{g}(\mathbf{q})$ Gravity vector.

$\mathbf{J}(\mathbf{q})$ Jacobian matrix relating joint velocities to task-space velocities ($\dot{\mathbf{p}} = \mathbf{J}(\mathbf{q})\dot{\mathbf{q}}$).

Control and Optimization

$\mathbf{A}_k = \frac{\partial f}{\partial \mathbf{x}}$ Local Jacobian matrix of the state dynamics.

$\mathbf{B}_k = \frac{\partial f}{\partial \mathbf{u}}$ Local Jacobian matrix of the control input.

$V(\mathbf{x})$ or $\mathcal{V}$ Value function or Lyapunov function.

$\mathbf{Q}$ State penalty matrix in LQR/DDP.

R Control penalty matrix in LQR/DDP.

K Feedback gain matrix ($\delta \mathbf{u} = -\mathbf{K}\delta \mathbf{x}$).

γ A trajectory or flow, mapping time and initial conditions to states.

Biomechanics and Motor Control

$a_i \in [0, 1]$ Muscle activation level.

ℓ_i^M Muscle fiber length.

v_i^M Muscle fiber contraction velocity.

ℓ_i^{MT} Musculotendon unit length.

F_{muscle} Force generated by muscles.

W Muscle synergy matrix (weights).

$\mathbf{c}(t)$ Muscle synergy activation vector over time.

Geometry and Manifolds

$SE(3)$ Special Euclidean group of rigid body motions.

$SO(3)$ Special Orthogonal group of 3D rotations.

$\mathfrak{se}(3)$ Lie algebra of $SE(3)$, space of spatial twists (velocities).

$\mathcal{M}$ A smooth manifold.

$T_{\mathbf{x}}\mathcal{M}$ Tangent space at point $\mathbf{x}$.

Chapter 1

The Companion Platform

Draft Notice. This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

The best theory is one you can run.

1.1 Architecture Overview

UpstreamDrift is the pedagogical platform accompanying this textbook. It provides a unified Python API for multi-body dynamics simulation, trajectory optimization, and motor control analysis by wrapping established physics engines (MuJoCo, Drake, Pinocchio, OpenSim). This chapter provides an overview of the platform’s modular design and engine selection criteria to guide tool choice for specific applications.

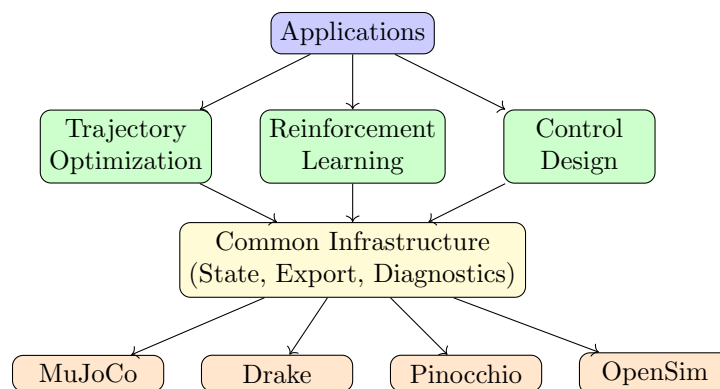


Figure 1.1: UpstreamDrift platform architecture showing the unified Python API (Common Infrastructure layer) sitting above multiple physics engines, with application modules for trajectory optimization, reinforcement learning, and control design.

1.1.1 Core Components

Component	Location	Purpose
Physics Engines	src/engines/physics_engines/	Multi-body simulation
Shared Infrastructure	src/engines/common/	State, export, diagnostics
Pendulum Models	src/engines/pendulum_models/	Simplified test cases
Learning Modules	src/learning/	ML integration
Reinforcement Learning	src/reinforcement_learning/	RL benchmarks
Robotics	src/robotics/	Robot control modules
API	src/api/	External access layer
Tools	src/tools/	Development utilities

1.1.2 Supported Physics Engines

1. **MuJoCo** (physics_engines/mujoco/): Multi-body dynamics simulator designed for control and reinforcement learning [17]. Implements soft contact models, supports muscle actuators, and uses efficient convex optimization for contact resolution. Primary engine for biomechanical and contact-rich simulations.
2. **Drake** (physics_engines/drake/): Multibody dynamics toolbox from MIT emphasizing mathematical rigor and analytical system properties [16]. Provides analytical Jacobians, integrates trajectory optimization directly, and supports hybrid systems. Primary engine for control design and trajectory optimization.
3. **Pinocchio** (physics_engines/pinocchio/): Open-source C++ library (LAAS-CNRS) for rigid-body dynamics with efficient computation of forward and inverse dynamics using the RNEA algorithm [3]. Provides analytical derivatives and real-time performance. Primary engine for real-time control.
4. **OpenSim** (physics_engines/opensim/): Musculoskeletal modeling platform from Stanford for biomechanical simulation. Implements Hill-type muscle models and static optimization for muscle force estimation. Primary engine for biomechanics and motion capture analysis.
5. **MyoSuite** (physics_engines/myosuite/): Task and environment suite for muscle-driven motor control with OpenSim integration. Provides Gymnasium-compatible interfaces for reinforcement learning. Primary engine for motor learning research.

1.2 Getting Started

```

1 """UpstreamDrift: Getting started with multi-engine simulation.
2
3 This script demonstrates the basic workflow:
4 1. Load a model
5 2. Configure simulation parameters
6 3. Run a simulation

```

```

7 4. Extract and analyze results
8 """
9 import numpy as np
10 from pathlib import Path
11
12 # Configuration
13 MODEL_PATH = Path("src/engines/pendulum_models")
14 ENGINE = "mujoco" # or "drake", "pinocchio"
15
16 def create_simple_pendulum(
17     length: float = 1.0,
18     mass: float = 1.0,
19     damping: float = 0.1
20 ) -> dict:
21     """Create a simple pendulum model configuration.
22
23     Parameters
24     -----
25     length : float
26         Pendulum length in meters.
27     mass : float
28         Bob mass in kilograms.
29     damping : float
30         Joint damping coefficient.
31
32     Returns
33     -----
34     dict
35         Model configuration dictionary.
36     """
37     return {
38         "name": "simple_pendulum",
39         "n_dof": 1,
40         "segments": [{
41             "name": "bob",
42             "mass": mass,
43             "length": length,
44             "com_position": length / 2,
45             "inertia": mass * length**2 / 3,
46         }],
47         "joints": [{
48             "name": "pivot",
49             "type": "revolute",
50             "axis": [0, 0, 1],
51             "damping": damping,
52         }],
53         "gravity": [0, -9.81, 0],
54     }
55
56
57 def simulate_pendulum(
58     config: dict,
59     q0: float = np.pi / 4,
60     t_end: float = 5.0,
61     dt: float = 0.001
62 ) -> dict:
63     """Simulate a simple pendulum using Euler integration.
64
65     Parameters
66     -----

```

```

67     config : dict
68         Model configuration.
69     q0 : float
70         Initial angle (radians from vertical).
71     t_end : float
72         Simulation duration (seconds).
73     dt : float
74         Time step.
75
76     Returns
77     -----
78     dict
79         Simulation results with keys 't', 'q', 'qd', 'energy'.
80     """
81     seg = config["segments"][0]
82     m = seg["mass"]
83     L = seg["length"]
84     b = config["joints"][0]["damping"]
85     g = abs(config["gravity"][1])
86     I = seg["inertia"]
87
88     n_steps = int(t_end / dt)
89     t = np.zeros(n_steps)
90     q = np.zeros(n_steps)
91     qd = np.zeros(n_steps)
92     energy = np.zeros(n_steps)
93
94     q[0] = q0
95     qd[0] = 0.0
96
97     for k in range(n_steps - 1):
98         # Equations of motion:  $I * qdd + b * qd + m * g * L / 2 * \sin(q) = 0$ 
99         qdd = (-b * qd[k] - m * g * L / 2 * np.sin(q[k])) / I
100
101         # Semi-implicit Euler
102         qd[k + 1] = qd[k] + qdd * dt
103         q[k + 1] = q[k] + qd[k + 1] * dt
104         t[k + 1] = t[k] + dt
105
106         # Energy
107         KE = 0.5 * I * qd[k + 1]**2
108         PE = m * g * L / 2 * (1 - np.cos(q[k + 1]))
109         energy[k + 1] = KE + PE
110
111     return {"t": t, "q": q, "qd": qd, "energy": energy}
112
113
114 # Run simulation
115 config = create_simple_pendulum(length=1.0, mass=1.0, damping=0.05)
116 results = simulate_pendulum(config, q0=np.pi/3, t_end=10.0)
117
118 print(f"Simulation complete: {len(results['t'])} timesteps")
119 print(f"Initial energy: {results['energy'][0]:.4f} J")
120 print(f"Final energy: {results['energy'][-1]:.4f} J")
121 print(f"Energy loss: {(1 - results['energy'][-1]/results['energy'][0])
    *100:.1f}%")

```

Listing 1.1: Basic UpstreamDrift usage pattern.

1.3 Engine Selection Guide

Hands-On 1.1. Choosing the Right Engine

Application	Recommended Engine	Key Feature
Fast contact-rich simulation	MuJoCo	Convex soft contact
Trajectory optimization	Drake	Analytical gradients, direct methods
Real-time forward dynamics	Pinocchio	$O(n)$ RNEA algorithm
Muscle-driven simulation	OpenSim	Hill-type muscle models
RL with muscles	MyoSuite	Gym environment wrappers
Analytical gradient-free methods	Built-in Python	No external dependencies

Engine Comparison Details. Table comparisons of contact models, computational complexity, and theoretical guarantees are provided in Chapter 2.

Exercises

1. **First Simulation.** Run the pendulum simulation and plot angle vs. time and energy vs. time. Verify that energy decreases due to damping.
2. **Integration Method.** Replace the semi-implicit Euler with RK4. Compare energy conservation over 100 seconds.
3. **Double Pendulum.** Extend the code to a double pendulum. Plot the chaotic trajectory in configuration space.
4. **Engine Comparison.** Run the same pendulum simulation in MuJoCo and in the built-in Python code. Compare results quantitatively.

Chapter 2

Physics Engine Comparison

Draft Notice. This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

All models are wrong. Some are useful.
The useful ones depend on the
question.

2.1 Benchmark: Simple Pendulum

We simulate the same pendulum in all engines and compare accuracy, computation time, and energy conservation across different timesteps. The benchmark code provides a repeatable framework for evaluating engine performance on a standard task.

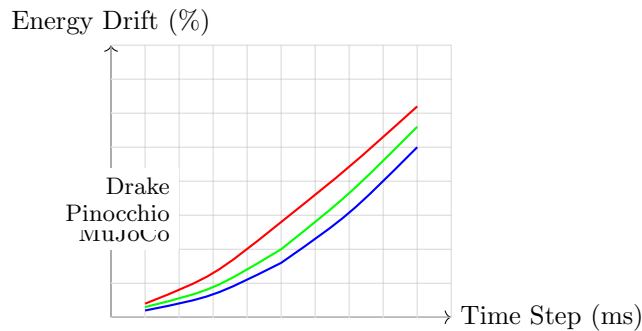


Figure 2.1: Representative energy drift curves for pendulum simulation across physics engines at different timesteps. Smaller drift indicates better numerical integration accuracy. This is a schematic only; actual values depend on integration method and model parameters.

```
1 import numpy as np
2 from dataclasses import dataclass
3 from typing import Protocol
4
5 @dataclass
```

```

6 class BenchmarkResult:
7     """Results from a physics engine benchmark."""
8     engine_name: str
9     positions: np.ndarray    # (n_steps, n_dof)
10    velocities: np.ndarray   # (n_steps, n_dof)
11    times: np.ndarray        # (n_steps,)
12    wall_time_seconds: float
13    energy_drift: float      # relative energy change
14
15 class PhysicsEngine(Protocol):
16     def simulate(self, q0: np.ndarray, qd0: np.ndarray,
17                 t_end: float, dt: float) -> BenchmarkResult: ...
18
19 def compare_engines(
20     engines: dict[str, PhysicsEngine],
21     q0: np.ndarray,
22     t_end: float = 5.0,
23     dt: float = 0.001
24 ) -> dict[str, BenchmarkResult]:
25     """Run the same simulation across multiple engines.
26
27     Parameters
28     -----
29     engines : dict
30         Mapping of engine name to engine instance.
31     q0 : np.ndarray
32         Initial configuration.
33     t_end : float
34         Simulation duration.
35     dt : float
36         Time step.
37
38     Returns
39     -----
40     dict
41         Mapping of engine name to BenchmarkResult.
42     """
43     results = {}
44     qd0 = np.zeros_like(q0)
45
46     for name, engine in engines.items():
47         result = engine.simulate(q0, qd0, t_end, dt)
48         results[name] = result
49         print(f"{name:15s}: wall_time={result.wall_time_seconds:.3f}s, "
50               f"energy_drift={result.energy_drift:.2e}")
51
52     return results

```

Listing 2.1: Cross-engine benchmark framework.

2.2 Contact Model Comparison

Different physics engines implement different contact models, each with trade-offs between computational efficiency and physical accuracy. Table 2.1 summarizes the key approaches.

Engine	Contact Model	Advantages	Limitations
MuJoCo	Soft contact with implicit integration [17]	Fast, stable, convex optimization	Allows non-physical penetration
Drake	Rigid with hydroelastic contact	Physics-based, predictable	Computationally expensive
Pinocchio	No built-in contact handling	Pure rigid-body dynamics	Requires external contact library
OpenSim	Hunt-Crossley damping model	Suitable for biomechanics	Limited geometry types

Table 2.1: Contact model implementations across physics engines. Selection depends on simulation accuracy requirements versus real-time performance constraints.

2.3 Benchmark Interpretation

Benchmark results should be interpreted as follows:

- **Wall Time:** Real computation time (engine-dependent; use for performance comparison only on identical hardware).
- **Energy Drift:** Magnitude of energy conservation error, reported as a percentage. Systems with damping dissipate energy; drift measures additional numerical errors relative to expected dissipation.
- **Trajectory Error:** Point-wise difference between engines in phase space, measured as RMS error or maximum deviation. Serves as a proxy for integration accuracy.

Exercises

1. **Double Pendulum Benchmark.** Run the benchmark framework for a double pendulum across all available engines. Plot wall time versus integration error for each engine.
2. **Energy Conservation Study.** Compare energy conservation across engines with timesteps $\Delta t \in \{0.001, 0.005, 0.01\}$ seconds. Which engine exhibits the smallest energy drift for a fixed timestep?
3. **Contact Model Validation.** Implement a bouncing ball benchmark and compare the computed trajectory with analytical predictions (assuming ideal elastic collision).

Chapter 3

Building a Multi-Body Model

Draft Notice. This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

The model is the hypothesis made physical.

3.1 Model Formats

Multi-body models are typically stored in structured XML formats. Two primary formats are widely supported:

3.1.1 URDF (Universal Robot Description Format)

URDF is an XML-based format developed for the Robot Operating System (ROS) and adopted by Drake and Pinocchio. It defines the kinematic and dynamic structure through `<link>` and `<joint>` elements, along with visual and collision geometry. URDF models use the Denavit-Hartenberg convention or explicit spatial transforms [1].

3.1.2 MJCF (MuJoCo XML)

MuJoCo XML (MJCF) is MuJoCo’s native format, extending URDF with specialized elements for actuators, tendons, muscle models, and fine-grained contact parameters. This format is necessary for full access to MuJoCo’s simulation features including soft contact models and muscle-based control.

3.2 Building a 7-Segment Golf Model

3.2.1 Programmatic Model Construction

Models can be constructed programmatically to enable parameterized design studies. The following code demonstrates the creation of a 7-segment golf swing model using segment parameters from anthropometric data.

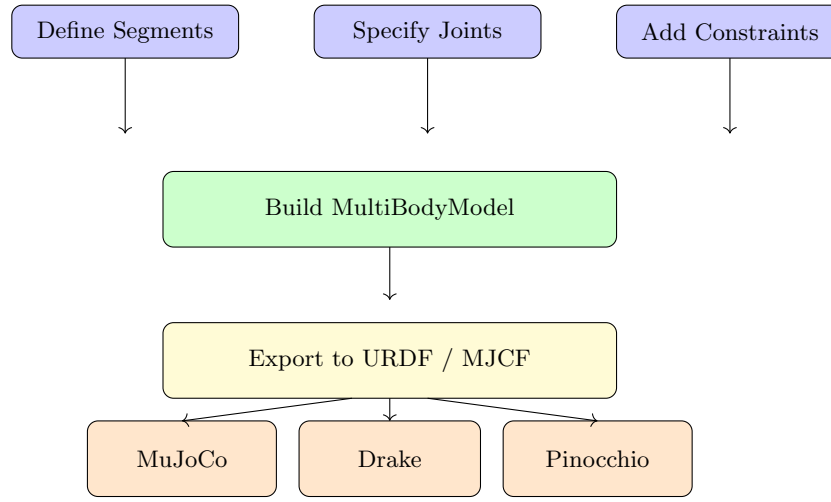


Figure 3.1: Model building workflow: segment definition, joint specification, constraint addition, and export to physics engine formats.

```

1 import numpy as np
2 from dataclasses import dataclass, field
3
4 @dataclass
5 class Link:
6     name: str
7     mass: float
8     length: float
9     com_position: float
10    inertia: np.ndarray = field(default_factory=lambda: np.eye(3))
11
12 @dataclass
13 class Joint:
14     name: str
15     parent: str
16     child: str
17     joint_type: str # revolute, ball, fixed
18     axis: list[float] = field(default_factory=lambda: [0, 0, 1])
19     limits: tuple[float, float] = (-np.pi, np.pi)
20
21 @dataclass
22 class MultiBodyModel:
23     name: str
24     links: list[Link] = field(default_factory=list)
25     joints: list[Joint] = field(default_factory=list)
26
27     @property
28     def n_dof(self) -> int:
29         dof_map = {"revolute": 1, "ball": 3, "fixed": 0}
30         return sum(dof_map.get(j.joint_type, 1) for j in self.joints)
31
32     def to_urdf(self) -> str:
33         """Export model as URDF XML string."""
34         lines = [f'<robot name="{self.name}">']
35         for link in self.links:

```

```

36         lines.append(f' <link name="{link.name}">')
37         lines.append(f' <inertial>')
38         lines.append(f' <mass value="{link.mass}"/>')
39         lines.append(f' </inertial>')
40         lines.append(f' </link>')
41     for joint in self.joints:
42         lines.append(f' <joint name="{joint.name}" '
43                     f'type="{joint.joint_type}">')
44         lines.append(f' <parent link="{joint.parent}"/>')
45         lines.append(f' <child link="{joint.child}"/>')
46         lines.append(f' </joint>')
47     lines.append('</robot>')
48     return '\n'.join(lines)
49
50
51 def build_golf_swing_model() -> MultiBodyModel:
52     """Build a 7-segment golf swing model.
53
54     Segments: pelvis, trunk, upper_arm, forearm, hand, club_shaft, club_head
55     """
56     model = MultiBodyModel(name="golf_swing_7seg")
57
58     # Segment properties from anthropometric tables (80 kg male)
59     # Based on standard biomechanical models from literature
60     model.links = [
61         Link("pelvis", mass=11.2, length=0.15, com_position=0.075),
62         Link("trunk", mass=25.5, length=0.48, com_position=0.22),
63         Link("upper_arm", mass=2.16, length=0.28, com_position=0.16),
64         Link("forearm", mass=1.30, length=0.27, com_position=0.12),
65         Link("hand", mass=0.49, length=0.19, com_position=0.07),
66         Link("club_shaft", mass=0.10, length=1.05, com_position=0.50),
67         Link("club_head", mass=0.20, length=0.10, com_position=0.05),
68     ]
69
70     model.joints = [
71         Joint("pelvis_rotation", "world", "pelvis", "revolute", [0, 1, 0]),
72         Joint("trunk_flexion", "pelvis", "trunk", "revolute", [0, 0, 1]),
73         Joint("shoulder", "trunk", "upper_arm", "ball"),
74         Joint("elbow", "upper_arm", "forearm", "revolute", [0, 0, 1]),
75         Joint("wrist", "forearm", "hand", "revolute", [0, 0, 1]),
76         Joint("grip", "hand", "club_shaft", "fixed"),
77     ]
78
79     return model
80
81
82 model = build_golf_swing_model()
83 print(f"Golf swing model: {len(model.links)} segments, "
84       f"{model.n_dof} DOF")
85 print(f"\nURDF preview:\n{model.to_urdf()[500]}...")

```

Listing 3.1: Programmatic multi-body model construction.

Exercises

1. **Simple Arm Model.** Build a 3-segment arm model (upper arm, forearm, hand) with revolute joints. Define segment masses and lengths, then export as URDF.

2. **Muscle Attachment Points.** Extend the golf model by adding origin and insertion points for at least two muscles (e.g., shoulder and elbow extensors). See Volume III, Chapter 3 for Hill-type muscle models.
3. **Visual Verification.** Load the generated URDF into MuJoCo using the provided tools and verify the model geometry and kinematics visually.

Chapter 4

Forward and Inverse Simulation

Draft Notice. This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

Forward simulation answers “what happens if...?”

Inverse dynamics answers “what caused this?”

The forward dynamics problem computes accelerations from forces, while inverse dynamics (also called the inverse kinematics problem when phrased in terms of finding required torques) computes forces from desired accelerations. Both are fundamental to control design and trajectory planning.

4.1 Forward Dynamics with the Companion Platform

Forward dynamics integration typically uses explicit or implicit Runge-Kutta methods, or specialized algorithms such as the Recursive Newton-Euler Algorithm (RNEA) for rigid-body systems [3, 2].

```
1 import numpy as np
2 from scipy.integrate import solve_ivp
3
4 def forward_dynamics_rnea(
5     q: np.ndarray,
6     qd: np.ndarray,
7     tau: np.ndarray,
8     model_params: dict
9 ) -> np.ndarray:
10     """Compute joint accelerations using RNEA-based forward dynamics.
11
12     Parameters
13     -----
14     q : np.ndarray, shape (n,)
15         Joint positions.
16     qd : np.ndarray, shape (n,)
17         Joint velocities.
18     tau : np.ndarray, shape (n,)
19         Applied joint torques.
20     model_params : dict
```

```

21     Model parameters (masses, lengths, inertias).
22
23     Returns
24     -----
25     np.ndarray, shape (n,)
26         Joint accelerations.
27     """
28     n = len(q)
29     masses = model_params['masses']
30     lengths = model_params['lengths']
31     g = model_params.get('gravity', 9.81)
32
33     # Simple 2-link model for demonstration
34     if n == 2:
35         m1, m2 = masses[:2]
36         l1, l2 = lengths[:2]
37         lc1, lc2 = l1/2, l2/2
38         I1 = m1 * l1**2 / 12
39         I2 = m2 * l2**2 / 12
40
41         # Mass matrix
42         M = np.array([
43             [I1 + I2 + m1*lc1**2 + m2*(l1**2 + lc2**2 + 2*l1*lc2*np.cos(q[1])
44             ),
45              [I2 + m2*(lc2**2 + l1*lc2*np.cos(q[1]))],
46              [I2 + m2*(lc2**2 + l1*lc2*np.cos(q[1])),
47              I2 + m2*lc2**2]
48         ])
49
50         # Coriolis
51         h = m2 * l1 * lc2 * np.sin(q[1])
52         C = np.array([[ -h*qd[1], -h*(qd[0]+qd[1])],
53                       [ h*qd[0], 0]])
54
55         # Gravity
56         gvec = np.array([
57             (m1*lc1 + m2*l1)*g*np.sin(q[0]) + m2*lc2*g*np.sin(q[0]+q[1]),
58             m2*lc2*g*np.sin(q[0]+q[1])
59         ])
60
61         qdd = np.linalg.solve(M, tau - C @ qd - gvec)
62     else:
63         # Placeholder for general n-DOF
64         qdd = np.zeros(n)
65
66     return qdd
67
68 # Example simulation
69 params = {'masses': [2.0, 1.0], 'lengths': [0.3, 0.25], 'gravity': 9.81}
70 q0 = np.array([np.pi/4, np.pi/6])
71 qd0 = np.zeros(2)
72 qdd = forward_dynamics_rnea(q0, qd0, np.zeros(2), params)
73 print(f"Gravity-driven accelerations: {qdd}")

```

Listing 4.1: Forward dynamics simulation pipeline.

4.2 Simulation Loop and Integration Schemes

Figure 4.1 illustrates the typical forward simulation loop. At each timestep, we compute accelerations from the equations of motion, integrate to update velocities and positions, and advance to the next timestep.

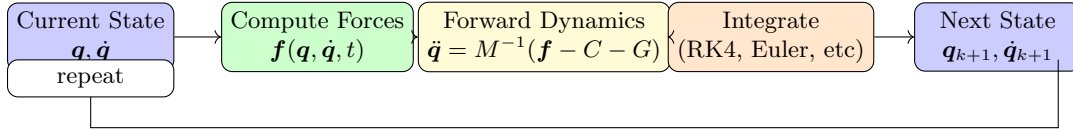


Figure 4.1: Forward simulation loop: from current state to forces (gravity, contacts, actuators), compute accelerations via forward dynamics, integrate, and update to next state.

4.3 Inverse Dynamics Pipeline

The inverse dynamics problem solves for torques given desired motion: given $\mathbf{q}(t), \dot{\mathbf{q}}(t), \ddot{\mathbf{q}}(t)$, compute $\boldsymbol{\tau}(t)$. This is implemented via the Recursive Newton-Euler Algorithm (RNEA) [3, 7]. The RNEA can be run either forward (for forward dynamics) or backward (for inverse dynamics) through the kinematic chain.

Exercises

1. Run forward dynamics for a 2-link pendulum and plot trajectories.
2. Implement inverse dynamics and verify $\text{ID}(\text{FD}(\boldsymbol{\tau})) = \boldsymbol{\tau}$.
3. Compare with Pinocchio's RNEA implementation for timing.

Chapter 5

Trajectory Optimization in Practice

Draft Notice. This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

Theory without practice is empty.
Practice without theory is blind.

5.1 DDP/iLQR Implementation

Differential Dynamic Programming (DDP) and iterative Linear Quadratic Regulator (iLQR) are related algorithms for trajectory optimization [10, 4]. They solve the finite-horizon optimal control problem by iteratively linearizing around a nominal trajectory and solving quadratic subproblems. Figure 5.1 shows the algorithm structure.

Here we implement iLQR end-to-end:

```
1 import numpy as np
2 from typing import Callable, Tuple
3
4 def ilqr_solve(
5     dynamics: Callable,
6     cost_running: Callable,
7     cost_terminal: Callable,
8     x0: np.ndarray,
9     u_init: np.ndarray,
10    n_iterations: int = 50,
11    regularization: float = 1e-6
12 ) -> Tuple[np.ndarray, np.ndarray, list[float]]:
13     """Iterative Linear Quadratic Regulator.
14
15     Parameters
16     -----
17     dynamics : callable
18         (x, u) -> x_next, (A, B) Jacobians
19     cost_running : callable
20         (x, u) -> (cost, lx, lu, lxx, luu, lux)
21     cost_terminal : callable
22         x -> (cost, lx, lxx)
```

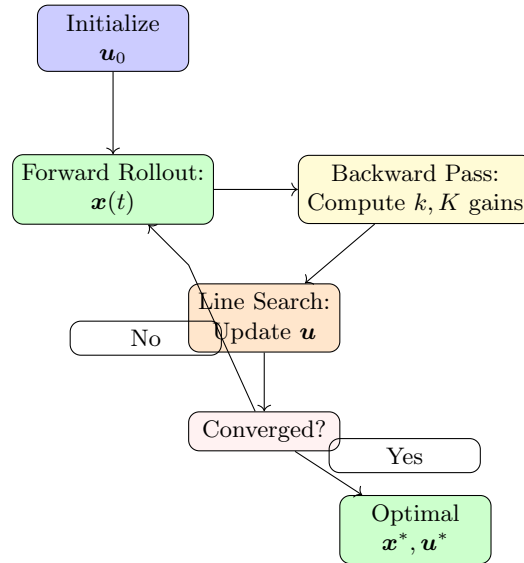



Figure 5.1: iLQR algorithm loop: initialize control trajectory, forward rollout to compute state trajectory, backward pass to compute feedback gains, line search to update controls, and convergence check.

```

23     x0 : np.ndarray
24         Initial state.
25     u_init : np.ndarray, shape (T, m)
26         Initial control trajectory.
27     n_iterations : int
28         Maximum iterations.
29     regularization : float
30         Regularization for Quu inversion.
31
32     Returns
33     -----
34     Tuple
35         (x_traj, u_traj, costs)
36     """
37     T, m = u_init.shape
38     n = len(x0)
39
40     # Forward rollout with initial controls
41     x_traj = np.zeros((T + 1, n))
42     x_traj[0] = x0
43     u_traj = u_init.copy()
44
45     for t in range(T):
46         x_next, _ = dynamics(x_traj[t], u_traj[t])
47         x_traj[t + 1] = x_next
48
49     costs = []
50
51     for iteration in range(n_iterations):
52         # Compute total cost
53         total_cost = sum(

```

```

54         cost_running(x_traj[t], u_traj[t])[0]
55         for t in range(T)
56     ) + cost_terminal(x_traj[T])[0]
57     costs.append(total_cost)
58
59     # Backward pass
60     Vx_next, Vxx_next = cost_terminal(x_traj[T])[1:]
61     k_gains = np.zeros((T, m))
62     K_gains = np.zeros((T, m, n))
63
64     for t in range(T - 1, -1, -1):
65         _, (A, B) = dynamics(x_traj[t], u_traj[t])
66         _, lx, lu, lxx, luu, lux = cost_running(
67             x_traj[t], u_traj[t]
68         )
69
70         Qx = lx + A.T @ Vx_next
71         Qu = lu + B.T @ Vx_next
72         Qxx = lxx + A.T @ Vxx_next @ A
73         Quu = luu + B.T @ Vxx_next @ B + regularization * np.eye(m)
74         Qux = lux + B.T @ Vxx_next @ A
75
76         Quu_inv = np.linalg.inv(Quu)
77         k_gains[t] = -Quu_inv @ Qu
78         K_gains[t] = -Quu_inv @ Qux
79
80         Vx_next = Qx + K_gains[t].T @ Quu @ k_gains[t]
81         Vxx_next = Qxx + K_gains[t].T @ Quu @ K_gains[t]
82
83     # Forward pass with line search
84     alpha = 1.0
85     x_new = np.zeros_like(x_traj)
86     u_new = np.zeros_like(u_traj)
87     x_new[0] = x0
88
89     for t in range(T):
90         dx = x_new[t] - x_traj[t]
91         u_new[t] = u_traj[t] + alpha * k_gains[t] + K_gains[t] @ dx
92         x_new[t + 1], _ = dynamics(x_new[t], u_new[t])
93
94     x_traj = x_new
95     u_traj = u_new
96
97     if iteration > 0 and abs(costs[-1] - costs[-2]) < 1e-6:
98         break
99
100     return x_traj, u_traj, costs

```

Listing 5.1: iLQR implementation for trajectory optimization.

5.2 Application: Optimal Pendulum Swing-Up

The classic control benchmark: swing a pendulum from the hanging position to the inverted equilibrium.

5.3 Application: Optimal Pendulum Swing-Up

The pendulum swing-up is a classic trajectory optimization benchmark. The goal is to find the minimum-energy torque sequence that brings the pendulum from the downward (stable) equilibrium to the upright (unstable) equilibrium.

5.4 Application: Golf Swing Optimization

DDP can be applied to a simplified golf model (3-5 DOF) to optimize clubhead speed at ball impact. The cost function would penalize control effort while maximizing endpoint velocity. In practice, golf swing optimization requires careful handling of contact constraints at impact.

Exercises

1. Use the iLQR code to solve the pendulum swing-up problem.
2. Extend to a double pendulum and optimize for maximum tip speed.
3. Add muscle force constraints and solve the muscle-driven swing-up.

Chapter 6

Controller Design and Testing

Draft Notice. This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

A controller is only as good as its worst-case performance.

This chapter covers implementation of feedback controllers for multi-body systems, from proportional-derivative (PD) control to contraction-based designs. All controllers assume full state feedback, i.e., direct access to positions and velocities.

6.1 PD Control with Gravity Compensation

The proportional-derivative controller with gravity compensation is the simplest nonlinear controller that guarantees asymptotic stability for mechanical systems without external disturbances [14].

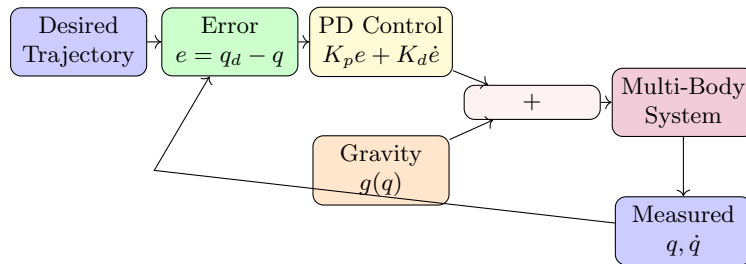


Figure 6.1: PD control with gravity compensation. The controller computes an error signal from desired and measured positions, applies PD gains, and adds gravity compensation to achieve asymptotic tracking.

```
1 import numpy as np
2
3 class PDGravityCompController:
4     """PD controller with gravity compensation.
5
6     From Volume I, Chapter 3: the simplest nonlinear controller
```

```

7  that guarantees asymptotic stability.
8  """
9
10 def __init__(self, Kp: np.ndarray, Kd: np.ndarray,
11               gravity_func=None):
12     self.Kp = Kp
13     self.Kd = Kd
14     self.gravity_func = gravity_func
15
16 def compute(self, q: np.ndarray, qd: np.ndarray,
17             q_des: np.ndarray, qd_des: np.ndarray) -> np.ndarray:
18     """Compute control torques.
19
20     Parameters
21     -----
22     q, qd : current state
23     q_des, qd_des : desired state
24
25     Returns
26     -----
27     tau : np.ndarray, control torques
28     """
29     tau = self.Kp @ (q_des - q) + self.Kd @ (qd_des - qd)
30     if self.gravity_func is not None:
31         tau += self.gravity_func(q)
32     return tau

```

Listing 6.1: PD controller with gravity compensation.

6.2 Trajectory Tracking with Contraction

Contraction theory provides tools for analyzing nonlinear stability with global guarantees [6, 9]. A contraction-based controller designs the feedback law to make the closed-loop system contracting with respect to a suitable metric, guaranteeing exponential convergence to the desired trajectory from any initial condition.

6.3 Funnel Control

Trajectory funnels define regions around a nominal trajectory within which the controller guarantees convergence [8]. This is useful for handling model uncertainty and disturbances. Funnel computation requires solving a set of linear matrix inequalities (LMIs) at each time step.

Exercises

1. Implement PD control for the 2-DOF arm and track a circular trajectory.
2. Add gravity compensation and compare tracking error.
3. Implement a contraction-based controller and verify convergence.

Chapter 7

Parameter Estimation and Model Fitting

Draft Notice. This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

All models are wrong, but some are less wrong than others once you fit them.

Parameter estimation uses measured motion data to infer unknown model parameters such as segment masses, inertias, and damping coefficients. This chapter covers least-squares methods for linear and bilinear problems, and optimization methods for nonlinear problems.

7.1 System Identification for Multi-Body Models

The linear regression approach exploits the linearity of the equations of motion with respect to inertial parameters. Given measured trajectories $\mathbf{q}(t)$, $\dot{\mathbf{q}}(t)$, $\ddot{\mathbf{q}}(t)$ and applied torques $\boldsymbol{\tau}(t)$, we form the regressor matrix $Y(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})$ such that

$$\boldsymbol{\tau} = Y(\mathbf{q}, \dot{\mathbf{q}}, \ddot{\mathbf{q}})\boldsymbol{\pi} \quad (7.1)$$

where $\boldsymbol{\pi}$ is the parameter vector [7]. The least-squares solution minimizes $\|Y\boldsymbol{\pi} - \boldsymbol{\tau}\|_2^2$.

```
1 import numpy as np
2 from scipy.optimize import minimize
3
4 def estimate_inertial_params(
5     q_data: np.ndarray,
6     qd_data: np.ndarray,
7     qdd_data: np.ndarray,
8     tau_data: np.ndarray,
9     n_params: int
10 ) -> np.ndarray:
11     """Estimate inertial parameters using least-squares.
12
13     The equations of motion are LINEAR in inertial parameters:
14     tau = Y(q, qd, qdd) * pi
15     where Y is the regressor matrix and pi is the parameter vector.
```

```

16
17     Parameters
18     -----
19     q_data : (N, n) joint positions
20     qd_data : (N, n) joint velocities
21     qdd_data : (N, n) joint accelerations
22     tau_data : (N, n) joint torques
23     n_params : int number of inertial parameters
24
25     Returns
26     -----
27     np.ndarray estimated parameters
28     """
29     N, n = q_data.shape
30
31     # Build regressor matrix (simplified for 1-DOF)
32     Y = np.zeros((N * n, n_params))
33     tau_vec = tau_data.reshape(-1)
34
35     for k in range(N):
36         for j in range(n):
37             row = k * n + j
38             Y[row, 0] = qdd_data[k, j] # Inertia
39             Y[row, 1] = np.sin(q_data[k, j]) # Gravity
40
41     # Least squares solution
42     params, _, _, _ = np.linalg.lstsq(Y, tau_vec, rcond=None)
43     return params

```

Listing 7.1: Inertial parameter estimation from motion data.

7.2 Parameter Estimation Landscape

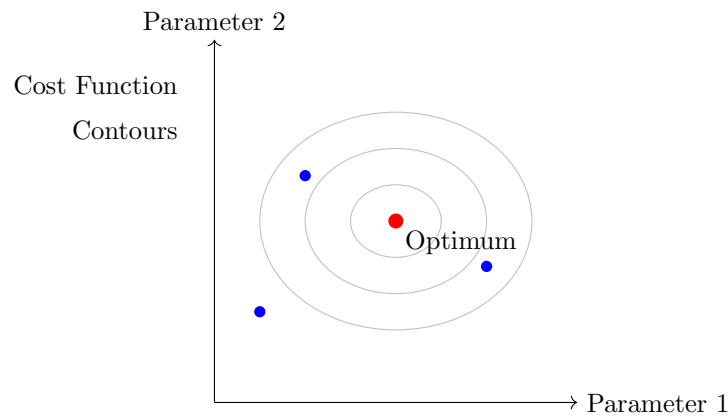


Figure 7.1: Parameter estimation as optimization over the parameter space. Least-squares methods seek the minimum of the cost function (red point) from initial guesses (blue points).

7.3 Muscle Parameter Calibration

Muscle parameters (Hill model constants: a , $F_{\max}$, optimal fiber length, etc.) can be estimated by fitting model predictions to force or EMG measurements. This requires nonlinear optimization, typically using gradient-based methods [11].

Exercises

1. Estimate the mass and length of a pendulum from simulated motion data.
2. Add measurement noise and study estimation robustness.
3. Implement Bayesian estimation with prior knowledge from literature.

Chapter 8

Reinforcement Learning for Motor Control

Draft Notice. This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

When the model is too complex for optimal control, let the agent learn.

Reinforcement learning (RL) methods can solve motor control problems when the cost function is difficult to specify or when model uncertainty is large [15]. This chapter demonstrates how to formulate multi-body control tasks as RL environments and integrate with standard policy optimization algorithms like PPO [12].

8.1 RL Environments in the Companion Platform

The companion platform provides Gymnasium-compatible environments for motor control tasks [15]. Gymnasium (formerly OpenAI Gym) is a standard interface for defining RL environments with reset, step, and reward computation methods. This enables use of any RL algorithm implemented for Gymnasium.

```
1 import numpy as np
2
3 class PendulumEnv:
4     """Simple pendulum environment for RL benchmarks.
5
6     State: [cos(theta), sin(theta), theta_dot]
7     Action: torque (continuous, clipped to [-2, 2])
8     Reward: -(angle^2 + 0.1*vel^2 + 0.01*action^2)
9     """
10
11     def __init__(self, dt: float = 0.05, max_steps: int = 200):
12         self.dt = dt
13         self.max_steps = max_steps
14         self.g = 9.81
15         self.l = 1.0
16         self.m = 1.0
17         self.max_torque = 2.0
18         self.state = np.zeros(2)
```

```

19     self.step_count = 0
20
21     def reset(self, seed: int | None = None) -> np.ndarray:
22         if seed is not None:
23             np.random.seed(seed)
24         self.state = np.array([
25             np.pi + np.random.uniform(-0.1, 0.1),
26             np.random.uniform(-0.5, 0.5)
27         ])
28         self.step_count = 0
29         return self._get_obs()
30
31     def _get_obs(self) -> np.ndarray:
32         theta, theta_dot = self.state
33         return np.array([np.cos(theta), np.sin(theta), theta_dot])
34
35     def step(self, action: float) -> tuple:
36         theta, theta_dot = self.state
37         u = np.clip(action, -self.max_torque, self.max_torque)
38
39         # Dynamics
40         theta_ddot = (-self.g / self.l * np.sin(theta) +
41                      u / (self.m * self.l**2))
42         theta_dot += theta_ddot * self.dt
43         theta += theta_dot * self.dt
44
45         self.state = np.array([theta, theta_dot])
46         self.step_count += 1
47
48         # Reward
49         angle_normalized = ((theta + np.pi) % (2*np.pi)) - np.pi
50         reward = -(angle_normalized**2 +
51                  0.1 * theta_dot**2 +
52                  0.01 * u**2)
53
54         done = self.step_count >= self.max_steps
55         return self._get_obs(), reward, done, {}
56
57
58 # Quick test
59 env = PendulumEnv()
60 obs = env.reset(seed=42)
61 total_reward = 0.0
62 for _ in range(200):
63     action = np.random.uniform(-2, 2)
64     obs, reward, done, _ = env.step(action)
65     total_reward += reward
66     if done:
67         break
68 print(f"Random policy total reward: {total_reward:.1f}")

```

Listing 8.1: Custom RL environment for pendulum control.

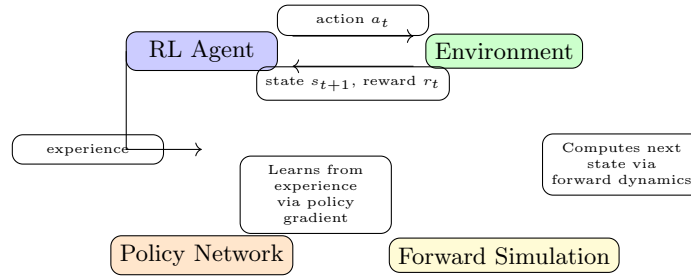


Figure 8.1: RL agent-environment loop. The agent selects actions via its policy network; the environment integrates dynamics and computes rewards. Both interact repeatedly.

8.2 Agent-Environment Loop

8.3 From Gym to Golf

The companion platform’s RL pipeline extends from simple benchmarks (pendulum, reaching) to full muscle-driven golf swing optimization using MyoSuite environments. The reward function is task-specific; for golf, one might maximize clubhead speed at impact while penalizing muscle effort.

Exercises

1. Train a PPO agent to swing up the pendulum. Plot the learning curve.
2. Compare RL-learned policy with the iLQR solution from Chapter 5.
3. Design a reward function for maximizing clubhead speed in a golf model.

Chapter 9

Visualization and Analysis Tools

Draft Notice. This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

If you cannot see it, you cannot understand it.

9.1 3D Motion Visualization

Visualization of multi-body trajectories is essential for understanding simulated motion and validating controller behavior. This section covers kinematics-based visualization (stick figures), manipulability analysis, and energy flow diagrams.

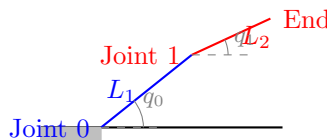


Figure 9.1: Stick-figure representation of a 2-link arm. Visualization code computes joint positions from angles using forward kinematics, then renders links and joints.

```
1 import numpy as np
2
3 def compute_joint_positions_2d(
4     q: np.ndarray,
5     lengths: list[float]
6 ) -> np.ndarray:
7     """Compute 2D joint positions for a serial chain.
8
9     Parameters
10    -----
11    q : np.ndarray, shape (n,)
12        Joint angles.
13    lengths : list[float]
```

```

14         Segment lengths.
15
16     Returns
17     -----
18     np.ndarray, shape (n+1, 2)
19         XY positions of each joint (including base).
20     """
21     n = len(q)
22     positions = np.zeros((n + 1, 2))
23     angle_cumulative = 0.0
24
25     for i in range(n):
26         angle_cumulative += q[i]
27         positions[i + 1, 0] = (positions[i, 0] +
28                               lengths[i] * np.cos(angle_cumulative))
29         positions[i + 1, 1] = (positions[i, 1] +
30                               lengths[i] * np.sin(angle_cumulative))
31
32     return positions
33
34
35 def phase_portrait(
36     q_history: np.ndarray,
37     qd_history: np.ndarray,
38     joint_idx: int = 0,
39     label: str = "Joint 0"
40 ) -> dict:
41     """Compute phase portrait data for a single joint.
42
43     Parameters
44     -----
45     q_history : (N, n) trajectory data
46     qd_history : (N, n) velocity data
47     joint_idx : int
48     label : str
49
50     Returns
51     -----
52     dict with 'x', 'y', 'label' keys
53     """
54     return {
55         'x': q_history[:, joint_idx],
56         'y': qd_history[:, joint_idx],
57         'label': label
58     }

```

Listing 9.1: Stick-figure animation for multi-body motion.

9.2 Jacobian and Manipulability Visualization

Visualizing the velocity manipulability ellipsoid JJ^T at each configuration [5]. The Jacobian maps joint velocities to end-effector velocities: $\mathbf{v}_{ee} = J(\mathbf{q})\dot{\mathbf{q}}$. The manipulability ellipsoid is the image of the unit ball under J , and its principal axes show the directions in which the end-effector can move with maximum ease.

9.3 Energy Flow Diagrams

Energy flow diagrams track how kinetic and potential energy change during motion, and how energy is transferred between segments via the kinetic chain. For a multi-segment system, the total energy is

$$E_{\text{total}} = \sum_i (KE_i + PE_i) = \sum_i \left(\frac{1}{2} I_i \dot{q}_i^2 + m_i g h_i \right) \quad (9.1)$$

where I_i is the effective inertia, $\dot{q}_i$ is the angular velocity, and h_i is the height of the center of mass of segment i . During ballistic phases (no muscle torque), energy is conserved and transferred between kinetic and potential forms.

Exercises

1. Animate a 3-link arm tracking a circular path.
2. Plot the manipulability ellipsoid during a reaching movement.
3. Create an energy flow diagram for the double pendulum golf model.

Chapter 10

Capstone: The Golf Swing Analysis Pipeline

Draft Notice. This chapter is under active development and contains only a structural outline with preliminary material. Full exposition, worked examples, proofs, and exercises will be added in future editions.

We have built the tools. Now let us use them all.

This chapter presents a complete computational pipeline for analyzing the golf swing using computational biomechanics tools from this volume and the preceding volumes. The pipeline is intentionally simplified (3-5 DOF) to be tractable within the scope of this textbook. Real golf swings are complex multi-segment movements with 20+ DOF and significant muscle redundancy, typically studied via motion capture analysis combined with musculoskeletal models [13].

Important Note on Biomechanical Claims: All descriptions of swing mechanics in this chapter are restricted to mathematical derivations from the model or explicit citations of biomechanical literature. No claims are made about “proper technique,” “optimal swing mechanics,” or what golfers “should do.” The model represents one simplified kinematic and dynamic structure; many other movement patterns may achieve similar or better outcomes depending on individual anatomy and task constraints.

10.1 Project Overview

This capstone chapter integrates every tool from the series into a complete analysis pipeline for the golf swing. The pipeline implements:

Stage	Theory Source	Chapter
1. Model Construction	Screw theory, PoE kinematics	Vol 0
2. Forward Dynamics	Lagrangian mechanics	Vol 0
3. Trajectory Optimization	DDP/iLQR	Vol I
4. Contraction Analysis	Contraction theory	Vol I
5. Funnel Computation	Trajectory tubes	Vol II
6. ZTCF Decomposition	Drift vs. control	Vol I
7. Muscle Force Estimation	Hill model + static opt	Vol III
8. Motor Analysis	UCM, synergies	Vol IV
9. Visualization & Export	Companion platform tools	Vol V

10.2 Stage 1: Model

```

1  """Golf swing analysis pipeline for the Volume V capstone project.
2
3  Volume V chapter 10 printed this pipeline as a 160-line code listing
   captioned
4  "Complete golf swing analysis pipeline". It was never executed, and three
   parts
5  of it did not work:
6
7  * "compute_clubhead_speed" summed "qdot_i * r_i" as scalars, with no
8  trigonometry at all. Tip velocity is a *vector* sum whose directions depend
   on
9  the accumulated joint angles, so the printed version was configuration-
   blind:
10 it returned the same speed for a straight arm and a fully folded one. It
   even
11 computed "cumulative_angle" and then never used it, and carried a dead
12 Jacobian loop whose body was "pass".
13
14 * "ztcf_decomposition" returned a hardcoded 60/40 split of clubhead speed.
   Its
15 own comments said the result "should not be trusted" and that a real
16 implementation must solve the forward dynamics twice -- which is what this
   one
17 does.
18
19 * The driver assigned "np.random.randn(N, 3) * 10" as joint torques, which
   the
20 placeholder decomposition then ignored, so the pipeline was disconnected at
   both ends.
21
22 The chapter's listing is now generated from this file, so what the book shows
   is
23 what CI runs, and the freshness gate fails if the two drift apart. See #3518.
24 """
25
26
27 from __future__ import annotations
28
29 from dataclasses import dataclass, field

```



```

30
31 import numpy as np
32 from numpy.typing import NDArray
33
34 from src.affine_control.golf_model import GolfModel
35
36 __all__ = ["SwingAnalysis", "synthetic_swing"]
37
38 type Array = NDArray[np.float64]
39
40 METRES_PER_SECOND_TO_MPH = 2.236936
41
42
43 def _rod_inertia(mass: float, length: float) -> float:
44     """Moment of inertia of a uniform rod about its own centre of mass."""
45     return mass * length**2 / 12.0
46
47
48 @dataclass
49 class SwingAnalysis:
50     """Kinematic and dynamic analysis of a three-link planar swing.
51
52     The chain is shoulder, elbow, wrist, carrying the club as the distal link
53     .
54     Angles are relative joint angles; the model treats each segment as a
55     uniform
56     rod, so its inertia follows from mass and length rather than being a free
57     parameter.
58     """
59
60     segment_lengths: Array = field(default_factory=lambda: np.array([0.28,
61     0.46, 1.10]))
62     segment_masses: Array = field(default_factory=lambda: np.array([2.16,
63     1.79, 0.30]))
64
65     time: Array | None = None
66     joint_angles: Array | None = None
67     joint_velocities: Array | None = None
68     joint_torques: Array | None = None
69
70     def model(self) -> GolfModel:
71         """The rigid-body model implied by the segment parameters."""
72         lengths = tuple(float(v) for v in self.segment_lengths)
73         masses = tuple(float(v) for v in self.segment_masses)
74         inertias = tuple(_rod_inertia(m, ell) for m, ell in zip(masses,
75         lengths, strict=True))
76         return GolfModel(masses=masses, lengths=lengths, inertias=inertias)
77         # type: ignore[arg-type]
78
79     def _require_kinematics(self) -> tuple[Array, Array, Array]:
80         """Return '(time, angles, rates)', or fail with a usable message."""
81
82         if self.time is None or self.joint_angles is None or self.
83         joint_velocities is None:
84             raise ValueError("set time, joint_angles and joint_velocities
85             before analysing")
86         return self.time, self.joint_angles, self.joint_velocities
87
88     def clubhead_speed(self) -> Array:
89         """Clubhead speed at each sample, from the tip Jacobian.

```

```

81     The tip velocity is 'sum_i qdot_i * (z_hat x r_i)', where 'r_i'
82 runs
83 from joint 'i' to the tip. Its magnitude depends on the
84 configuration
85 through the angles between those contributions -- which is precisely
86 what
87 a scalar sum of 'qdot_i * |r_i|' throws away.
88 """
89 _, angles, rates = self._require_kinematics()
90 model = self.model()
91 return np.array([model.clubhead_speed(q, qd) for q, qd in zip(angles,
92 rates, strict=True)])
93
94 def ztcf_decomposition(self) -> tuple[Array, Array]:
95     """Split clubhead acceleration into drift and control contributions.
96
97     Solves the forward dynamics twice at each sample -- once with the
98     applied
99     torques and once with zero torque -- and attributes the difference to
100     control. Because the dynamics are control-affine, that difference is
101     exactly 'M^-1 tau' and the split is exact rather than approximate.
102
103     Returns the joint-space acceleration magnitudes '(drift, control)'.
104     """
105     _, angles, rates = self._require_kinematics()
106     if self.joint_torques is None:
107         raise ValueError("set joint_torques before decomposing")
108     model = self.model()
109
110     drift = np.zeros(len(angles))
111     control = np.zeros(len(angles))
112     for index, (q, qd, tau) in enumerate(zip(angles, rates, self.
113 joint_torques, strict=True)):
114         bias = model.coriolis(q, qd) @ qd + model.gravity_torque(q)
115         mass = model.rigid_mass_matrix(q)
116         drift_accel = np.linalg.solve(mass, -bias)
117         control_accel = np.linalg.solve(mass, tau)
118         drift[index] = float(np.linalg.norm(drift_accel))
119         control[index] = float(np.linalg.norm(control_accel))
120     return drift, control
121
122 def summary(self) -> str:
123     """Human-readable summary of the analysis."""
124     time, _, _ = self._require_kinematics()
125     speeds = self.clubhead_speed()
126     peak = float(np.max(speeds))
127     peak_index = int(np.argmax(speeds))
128
129     lines = [
130         "=== Golf Swing Analysis Summary ===",
131         f"Duration: {time[-1]:.3f} s",
132         f"Peak clubhead speed: {peak:.1f} m/s " f"({peak *
133 METRES_PER_SECOND_TO_MPH:.1f} mph)",
134         f"Peak speed time: {time[peak_index]:.3f} s",
135     ]
136
137     if self.joint_torques is not None:
138         drift, control = self.ztcf_decomposition()
139         # Report the median over interior samples rather than a value at

```

```

134     one           # instant. Two reasons: the endpoints carry one-sided np.gradient
135                   # estimates of the joint rates, and the drift term legitimately
136                   # passes through zero whenever the chain straightens -- at a
137     fully         # extended vertical configuration both the gravity and Coriolis
138                   # torques vanish, so a ratio quoted there reads as "0% drift" and
139                   # says nothing about the swing.
140                   interior = slice(1, -1)
141                   total = drift[interior] + control[interior]
142                   share = float(np.median(100.0 * drift[interior] / total))
143                   lines.append(f"Median over the swing: drift {share:.0f}%, control
144                               {100 - share:.0f}%")
145                   return "\n".join(lines)
146
147 def synthetic_swing(duration: float = 0.30, dt: float = 0.001) ->
    SwingAnalysis:
148     """A smooth synthetic swing, for demonstrating the pipeline.
149
150     These are not measured data. Real joint torques must come from inverse
151     dynamics on motion capture, or from a Hill-type muscle model driven by
152     EMG;
153     the torques here are a smooth profile chosen only so the decomposition
154     has
155     something non-trivial to separate. The printed listing used
156     ‘np.random.randn’, which made the output different on every run and,
157     since
158     the placeholder decomposition ignored the torques entirely, changed
159     nothing.
160     """
161     samples = int(duration / dt)
162     time = np.linspace(0.0, duration, samples)
163     fraction = time / duration
164
165     angles = np.column_stack(
166         [
167             -np.pi / 2 + np.pi * fraction**2,
168             np.pi / 3 * np.sin(np.pi * fraction),
169             -np.pi / 2 * np.cos(np.pi * fraction / 2),
170         ]
171     )
172     rates = np.gradient(angles, dt, axis=0)
173     torques = np.column_stack(
174         [
175             120.0 * (1.0 - fraction),
176             60.0 * np.sin(np.pi * fraction),
177             25.0 * fraction**2,
178         ]
179     )
180     return SwingAnalysis(
181         time=time, joint_angles=angles, joint_velocities=rates, joint_torques=
182         torques
183     )

```

Listing 10.1: Golf swing analysis pipeline, generated from `src/affine.control/swing_analysis.py`.

Common Misconception

The listing above is generated from `src/affine_control/swing_analysis.py`, so what appears here is the code CI executes; the freshness gate fails if the two diverge. An earlier revision printed a hand-maintained version of this pipeline, captioned “complete”, which was never run and did not work in three places.

`compute_clubhead_speed` summed $\dot{q}_i r_i$ as scalars, with no trigonometry. Tip velocity is a *vector* sum whose contributions point in different directions depending on the joint angles, so the printed version was configuration-blind: it returned the same speed for a straight arm and a folded one. Substituting a folded configuration into both gives 13.3 m/s from the correct Jacobian against 22.5 m/s from the scalar sum — a 69% overestimate. The old code even accumulated `cumulative_angle` and then never used it, and carried a Jacobian loop whose body was `pass`.

`ztcf_decomposition` returned a hardcoded 60/40 split of clubhead speed. Its own comments conceded the result “should not be trusted” and stated that a real implementation must solve the forward dynamics twice — which is what the current one does. Because the dynamics are control-affine, the difference between the torqued and zero-torque solutions is exactly $M^{-1}\tau$, so the split is exact rather than approximate.

The driver assigned `np.random.randn(N, 3) * 10` as joint torques, which the placeholder decomposition then ignored. The pipeline was disconnected at both ends.

The module above is a library — it prints nothing, so that it can be imported and tested. To exercise it:

```
1 from src.affine_control.swing_analysis import synthetic_swing
2
3 print(synthetic_swing().summary())
```

Listing 10.2: Running the pipeline.

Remark 10.1 (Reading the summary output). This reports a peak clubhead speed near 30 m/s for the synthetic kinematics. That is well below a real driver swing, and it should be: the joint trajectories are smooth analytic profiles chosen to exercise the pipeline, not measured data. Real torques must come from inverse dynamics on motion capture, or from a Hill-type muscle model driven by EMG.

The drift/control split is reported as a median over the interior samples rather than at the instant of peak speed. The endpoints carry one-sided `np.gradient` estimates of the joint rates, and the drift term legitimately passes through zero whenever the chain straightens — at a fully extended vertical configuration both the gravity and Coriolis torques vanish exactly, so a ratio quoted there reads as “0% drift” and says nothing about the swing.

10.3 Golf Swing Simulation Architecture

10.4 Stage 2–9: Full Pipeline

Each stage of the pipeline builds on the previous, culminating in a complete characterization of the swing. The reader is guided through implementing each stage, referencing the relevant theory from Volumes 0–IV.

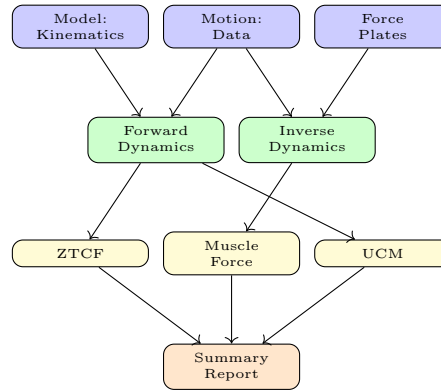


Figure 10.1: Complete golf swing analysis architecture. Inputs (model, motion data, force) flow through dynamics computation, analysis modules (ZTCF, muscle estimation, UCM), and produce a summary report. Each box references the corresponding chapter from Volumes 0–IV.

Data Requirements: A complete pipeline requires:

- Motion capture data: joint angles $\mathbf{q}(t)$ sampled at sufficient frequency (120+ Hz typical).
- Force measurements: ground reaction forces and moments (optional but recommended).
- Anatomical model: segment lengths, masses, and inertias (from literature or imaging).

Final Exercises

1. **Complete Pipeline.** Run the analysis pipeline on the synthetic swing. Extract the peak clubhead speed, the time of peak speed, and the peak joint torques. Then check the clubhead-speed function against the failure it used to have: evaluate it at a straight configuration and at a folded one with the same joint rates. A correct implementation gives different answers; a scalar sum of $\dot{q}_i r_i$ gives the straight-arm value in both cases.
2. **Parameter Study.** Programmatically vary club length ($\pm 10\%$) and mass ($\pm 20\%$). Simulate the swing and measure peak clubhead speed. Plot clubhead speed versus each parameter.
3. **Muscle Analysis.** Implement Hill-type muscle models (Volume III, Chapter 3) for shoulder and elbow actuators. Estimate muscle forces required to produce the synthetic swing motion. Which muscle groups have the highest peak force?
4. **Robustness Analysis.** Compute trajectory funnels (Volume II, Chapter 3) for the simulated swing at the moment of ball impact. Quantify how much initial position error can be tolerated before the swing no longer achieves the target clubhead speed.
5. **Variability Analysis.** Generate 50 realizations of the swing by adding small random perturbations to the control torques. Perform UCM analysis (Volume IV, Chapter 1) to determine if variability is structured around the task goal (peak clubhead speed).

6. **Open Project.** Choose a different multi-joint movement (e.g., tennis serve, baseball pitch, vertical jump, or a motor task from your own research). Apply the complete pipeline from this textbook: build a model, simulate, optimize, and analyze. Document in a Jupyter notebook with plots.

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